

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****May-June 2017****EC310.01/EC310 Optical Communication****Date: 03.06.2017, Saturday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [07]**

- a. Define Intermodal Dispersion. [01]
- b. Give the importance of Refractive Index. [01]
- c. Define Chromatic dispersion. [01]
- d. Define Population Inversion. [01]
- e. Give the advantage of fusion splicing. [01]
- f. What is group velocity? [01]
- g. Which among the following misalignments give/gives rise to the occurrence of splice loss? [01]
 - a. Longitudinal separation between the end-faces of fiber
 - b. Angular tilt between fiber ends
 - c. Transverse offset between fiber ends
 - d. All of the above

Q – 2.a Draw & Explain three types of mechanical misalignments. [04]**Q – 2.b Answer any two questions. [10]**

- 1) Derive the equation for the Numerical Aperture.
- 2) Define Bending loss and also explain types of bending loss occur in single mode fiber.
- 3) Discuss the key elements of optical fiber communication system.

Q - 3 Answer the following.

- a. Explain two popular alignment schemes used in fiber optic connectors. [05]
- b.
 - 1) Explain Fabry Perot resonator cavity for LASER diode. [05]
 - 2) What is internal quantum efficiency? Derive equation of internally generated power to LED. [04]

OR

- 1) Explain modulation of LED. [05]
- 2) Explain three key transition processes involved in laser action. [04]

SECTION – II

Q - 4 Answer the question below. [07]

- a. Define Junction Capacitance. [01]
- b. What is Impact ionization? [01]
- c. What is the advantage of Optical Amplifier? [01]
- d. Why we use DWDM? [01]
- e. Define splitting ratio in case of optical coupler. [01]
- f. What is switching? Give the Difference between Packet switching and Brust switching. [02]

Q – 5.a Which are the various types of noise occurs in photodiode and why? [05]

OR

Q – 5.a Explain principal of operation, the structure and the characteristic of Avalanche Photo Diode. [05]

Q – 5.b Explain the concept of WDM with neat diagram. [04]

Q – 5.c Explain 2 x 2 fiber coupler. What are the different losses associated with it? [05]

OR

Q – 5.c What is the operating principle of mach – zehnder interferometer? Draw the layout of it and explain in brief. [05]

Q – 6.a Write a short note on 4 Fiber BPSR. [05]

Q – 6.b Compare silicon RAPD with PIN photo detector with respect to the structure, principle of operation and the performance. State the advantages of RAPD. [05]

Q – 6.c What are Link Power Budget and Rise Time Budget? List out the parameter which will directly affect the rise time budget. [04]

OR

Q – 6.a A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900 nm. Suppose 0.5 μ W of optical power produces a multiplied photocurrent of 10 μ A. Find the multiplication M. ($h = 6.625 \times 10^{-34}$ JS) [04]

Q – 6.b Explain SONET-SDH Frame with neat sketch. [05]

Q – 6.c There are two systems available. Decide which can be used for longer distance transmission. **[05]**

System 1 (Operating at 850 nm)	System 2 (Operating at 1300 nm)
a) GaAlAs Laser Diode: 0 dBm fiber coupled power	a) InGaAsP LED : -13 dBm fiber coupled power
b) APD : -50 dBm Sensitivity	b) PIN diode: -38 dBm Sensitivity
c) Graded Index fiber : 3.5 dB/km attenuation at 850 nm	c) Graded Index fiber : 1.5 dB/km attenuation at 1300 nm
d) connector loss: 1dB/Connector	d) connector loss: 1dB/Connector

Allow 6db system margin in each case.
